Modelling and Evaluation of One-to-Three-Day Discharge and Flood-Risk Forecasts for the Huong River at Kim Long

Dang Nguyen Giap Ho Anh Duc Mai Thi Le Huyen

Department of Information Technology, FPT University
DSP391m — Data Science Project — Fall 2026
Hue, Vietnam

***Abstract*—**This report presents the modelling and evaluation stage of a project that forecasts the daily discharge and flood-risk level of the Huong River at Kim Long, Vietnam, one to three days ahead from open data. A LightGBM model on log discharge reaches NSE = 0.747, 0.420 and 0.175 at one, two and three days on a test period from July 2022, against 0.542, 0.008 and –0.198 for persistence, and a dedicated classifier detects the first risk level one day ahead with POD = 0.645 and FAR = 0.200. Responding to the review of Report 2, we tested reservoir releases, a longer record, large-scale weather drivers, finer and hourly satellite rainfall, and deep learning. GloFAS v4 shows no trace of the two upstream dams, and reported natural inflows exceed the simulated discharge on 14 of 23 days, so reservoir orders enter as a raise-only floor rule. Storm, cold-surge and ENSO predictors raise NSE at two days from 0.432 to 0.499 in the main model, which uses an L1 loss at one day and an L2 loss at two and three days, a choice made on the yearly walk-forward and the validation period, not on the test set. In a 13-fold yearly walk-forward, gauge-calibrated GSMaP rainfall adds +0.018 NSE at three days (positive in 11 of 13 folds), but the gain vanishes (–0.001) once the one-day delivery delay of the archive is imposed: it is a reanalysis result, not yet an operational one. LSTM and MLP models do not beat LightGBM consistently. SHAP attributions are plausible at the core (today's discharge and upstream rain, with pressure gradients gaining weight with lead time) but reveal a reversed soil-moisture effect. We also correct the catchment area used earlier from 2,830 to about 1,540 km².

***Keywords*—**flood forecasting, LightGBM, walk-forward validation, satellite precipitation, reservoir operation, GloFAS, rare-event verification, Huong River

I. INTRODUCTION

Reports 1 and 2 defined the task and the data: forecast the daily discharge of the Huong River at Kim Long (Hue, Vietnam) one, two and three days ahead, at the GloFAS v4 cell that represents the station (16.45° N, 107.50° E), and translate the forecast into flood-risk levels using open data only. This report covers model development and evaluation. The project is positioned as a local post-processing layer on top of the global forecast, in the line of recent GloFAS bias-correction studies [7], [8], not as a replacement for it.

TABLE I
FEEDBACK ON REPORT 2 AND THE RESPONSE IN THIS REPORT

Feedback	Work done	Finding
Reservoir releases, other factors	Dam-era test of GloFAS (R1); 132 reported dam flows (R2); soil moisture, storage proxies, storms, cold surges, ENSO	GloFAS v4 has no dams: releases enter a raise-only floor rule; weather drivers help at h = 2–3
Finer grid	30 m DEM sub-basins; satellite rain at 0.1°; PERSIANN-CCS at 0.04°	0.04° product misses monsoon rain; sub-basins add little
Hourly time step	54 hourly ERA5 features; hourly IMERG, GSMaP 1998–2026	Gain at h = 3 only, and only with same-day satellite data
Spark, deep learning (Report 4)	Two-worker Spark cluster; LSTM, MLP under walk-forward	Spark matches Polars but is slower; DL does not beat LightGBM

The review of Report 2 raised three points: river flow at Kim Long also depends on reservoir releases and other factors; the 5 km grid is coarse; and a daily time step may hide the rapid response of a steep basin. Distributed processing with Spark and, if possible, deep learning were requested for Report 4. Table I maps each point to the work done and the answer obtained. One constraint frames every answer: the target remains GloFAS daily discharge on a ~5 km grid, so finer or hourly *inputs* cannot make the *target* finer. Section II also corrects the catchment area used in Reports 1 and 2.

II. DATA UPDATES

A. Longer record and new predictors

ERA5 rainfall was extended back to 1997, the first year in which GloFAS returns discharge for this basin, which doubles the training set of the main split from 4,550 to 9,298 days. Four predictor families were added: ERA5-Land soil moisture in three layers; tropical cyclones from IBTrACS v04r01 [15] (417 storms within 1,500 km of Hue since 1996), described by distance, wind speed and approach speed; cold-surge indicators from ERA5 pressure and wind (the Hanoi–Hue pressure difference and the north-easterly wind); and the Oceanic Niño Index [16], lagged three months so that only values already published are used.

B. Correction to Report 2: catchment area

Report 1 gave the Huong catchment as about 2,830 km², and the grid-cell selection described there (and repeated in the Report 2 presentation) used that figure when checking the catchment area implied by specific runoff. The value is wrong for Kim Long. 2,830 km² is the whole Huong system *including* the Bo River (938 km²) [12], and the Bo joins downstream of Kim Long; the earlier reference for the combined system (3,768 km²) counted the Bo twice. Delineation from the Copernicus GLO-30 DEM [17] gives 1,543 km² to Kim Long, and the project now uses about 1,540 km². The selected cell does not change: its mean discharge of 120.7 m³/s implies 1,508–2,011 km² for a specific runoff of 0.060–0.080 m³/s per km², consistent with the corrected area. All area-based

quantities in this report use the corrected value.

C. Sub-basins from a 30 m DEM

Rainfall was previously averaged over three latitude bands. The catchment is now split into five non-overlapping sub-basins with a D8 flow model (Fig. 1): the Ta Trach and Binh Dien reservoir catchments (670 and 507 km², within –6.5% and –1.6% of the official 717 and 515 km² [11]), the Ta Trach and Huu Trach reaches below the dams (88 and 185 km²) and the reach from the Tuan confluence to Kim Long (93 km²). The 64 ERA5 cells cover only 92.5% of the catchment because its southern rim lies outside the rainfall box; the satellite grids below cover all of it.

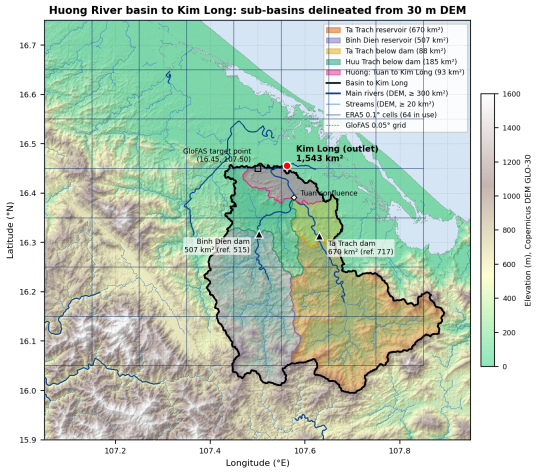


Fig. 1. Huong basin to Kim Long (1,543 km²), five sub-basins delineated from the 30 m DEM, ERA5 cells and the GloFAS target point.

D. Satellite rainfall

Hourly IMERG V07 [13] and GSMaP v8 [14] (satellite-only and gauge-calibrated, GC) were retrieved from Google Earth Engine for January 1998 to October 2026 on 100 cells of 0.1°. Only PERSIANN-CCS is finer than the GloFAS grid (0.04°), but for November 2020 it recorded 0.073 of the ERA5 total, probably because its cloud-top retrieval misses rain from the low, warm clouds of north-east monsoon events, so it was not used. For each basin and day nine hourly features were derived: daily total; maximum 1-, 3- and 6-h rain; hours with at least 1 mm; peak hour as sine and cosine; 24-h total to 07:00; and 72-h total. Over the 2015–2025 flood seasons IMERG, GSMaP and GSMaP-GC total 0.90, 0.74 and 1.14 times ERA5. ERA5 correlates best with next-day GloFAS discharge (r = 0.78 against 0.69 for IMERG and GSMaP-GC), as expected, because GloFAS v4 is itself driven by ERA5; this does not show that ERA5 is closer to the true rainfall.

E. Reservoir records

Release and inflow figures for the Ta Trach and Binh Dien reservoirs were collected from provincial bulletins and news: 132 records on 57 days from 2016 to 2026, each with source URL, access date and a verbatim quotation. No daily release series is published, so these records can test the target and drive an operating rule, but cannot be a training feature.

III. METHODOLOGY

A. Forecasting task

The target is the GloFAS v4 daily mean discharge at the Kim Long cell, h = 1, 2 and 3 days after the issue date t. Every predictor is known at the end of day t: rolling windows look backward only and an automated test fails if a feature uses later information. The main test period starts on 1 July 2022, when the Open-Meteo GloFAS series switches from reanalysis to archived forecasts (n = 1,522 days at h = 1).

B. Feature configurations

Configuration A is the Report 2 panel (66 features): discharge lags of 1–14 days, rainfall per band with lags and 3-, 5- and 7-day sums, an antecedent precipitation index (k = 0.9) and a cyclic day-of-year code. B–E add data cumulatively: B trains from 1997 with the same features; C adds soil moisture (70); D adds reservoir-storage proxies, namely 15- and 30-day rainfall and rainfall accumulated since 1 September, the start of the flood-operation season set by the inter-reservoir procedure [11] (73); E adds storm, cold-surge and ENSO predictors (88). Satellite features are tested on top of E (called E0 in the walk-forward runs) and of A (A0).

C. Models

The main model is LightGBM trained on log(1 + Q) (600 trees, learning rate 0.05, 31 leaves), one model per horizon; results are for seed 42 or, where stated, the mean of seeds 42, 7 and 2026. Report 2 fixed an L1 objective before any test result because it is robust to extremes, and the Report 2 model keeps it. For the main model the objective is L1 at h = 1 and L2 at h = 2 and 3. The switch was selected with the yearly walk-forward of Section III-E and the validation period, never with the test set: L2 raised the mean NSE by +0.039 [+0.020; +0.057] at h = 2 and +0.036 [+0.020; +0.051] at h = 3 (folds won/lost 11/2 at both horizons, Wilcoxon p = 0.003 and 0.002), and by +0.059 and +0.053 on validation; at h = 1 the fold gain (+0.018, p = 0.22) was not significant and the validation change was negative (–0.011), so L1 was kept. Configurations A–E all use this per-horizon rule, so the ablation compares predictors only. Neither loss removes the negative peak bias, so under-predicting peaks is a property of central regression on a skewed target. Risk levels are predicted by separate binary LightGBM classifiers per level and horizon, with class weights equal to the negative-to-positive ratio instead of resampling, which would break the temporal structure, and a decision threshold of 0.5; a threshold tuned on validation did not transfer to the test period and was not adopted.

Two neural families were compared (8 configurations, 3 seeds each): a multi-output LSTM reading a 14- or 30-day window of raw variables (discharge, band rainfall, API, season), optionally predicting the change from today's discharge and optionally with a peak-weighted loss; and an MLP on the LightGBM lag columns. Global LSTM models are

the state of the art for ungauged flood prediction [6], and transfer to small local basins is an open question [10].

D. Baselines

Persistence, climatology (multi-year mean of the calendar day, fitted on training data only), seasonal naive and ARIMA(2,1,2) on log discharge. The raw GloFAS forecast, the natural reference for a post-processing model, cannot be rebuilt for past dates: Open-Meteo archives past rainfall forecasts but not past flood forecasts (the historical flood endpoint returns HTTP 404), and using the GloFAS analysis at t + h as an input would leak the target. The issued GloFAS forecast has therefore been logged every morning since 15 September 2026 (one day, 20 September, was lost to a time-zone bug and is not imputed); the comparison will be reported in Report 4 on that operational sample.

E. Validation protocol

Single regime split. Train to December 2015, validation to June 2022, test from July 2022; final models are refit on train plus validation. **Yearly walk-forward.** Each year 2013–2025 is a test fold (13 folds); the training window expands from the start of the record; a 3-day gap separates train and test so that no h = 3 label enters the test year; and the risk threshold is recomputed from each fold's training data only. Gains are summarised by the mean ΔNSE over folds with a 95% bootstrap interval that resamples folds, the folds won and lost, and a two-sided Wilcoxon signed-rank test. January–August 2026 is excluded as a partial year.

Two facts bound the interpretation. Perturbing the input panel at the 13th decimal changes LightGBM NSE by about ±0.01, so smaller differences are not resolvable. And the archived-forecast regime contains only 3 full folds (2023–2025), for which the smallest attainable two-sided p is 0.25: that regime cannot be tested on its own.

F. Risk levels instead of alert stages

Official alert stages at the Hue (Kim Long) station are water levels of 1.0, 2.0 and 3.5 m [4], whereas the model forecasts discharge. An event-calibrated mapping from discharge to level was built from published flood peaks and failed: across 12 floods since 2010 log discharge explains little of the peak level (R^2 = 0.18), and the best fit on 7 floods (R^2 = 0.629) missed the five floods after July 2022 with a mean bias of –1.94 m. Tide and backwater from the Tam Giang lagoon, the Thao Long barrage, reservoir operation and the under-simulation of extreme floods all decouple level from simulated discharge. Labels are therefore *risk levels* at the 95th, 98th and 99.5th percentiles of September–December discharge on train plus validation: 806, 1,275 and 1,905 m³/s. They are relative to the GloFAS climate and are not alert stages.

G. Rare-event verification

Level-1 days are about 1% of the test period, so accuracy is never reported. Detection is scored with POD, FAR, CSI, F1, PR-AUC and the Brier score. A level supports a main conclusion only if the test set holds at least 30 positive days: level 1 has 31, level 2 only 12 and level 3 7; level 3 has just 9 positive days in train plus validation and is not modelled.

H. Reservoir floor rule

When the province orders releases, the forecast at Kim Long becomes the larger of the model forecast and the sum of the ordered releases at Ta Trach and Binh Dien, since released water reaches Kim Long within hours. The rule only raises a forecast and never lowers it; without an order nothing changes. Huong Dien and A Luoi are ignored because they drain to the Bo River or out of the basin. The daily job keeps the raw GloFAS value beside the floored one. The rule is a safeguard, not a validated component: no historical series of orders exists to score it.

IV. RESULTS

A. Baselines and LightGBM

LightGBM beats every baseline at every horizon (Table II). With the Report 2 predictors (configuration A, the reference for the acceptance criteria), its NSE at h = 1 of 0.747 meets the threshold of 0.70 and its RMSE is 25.6% below persistence (target 20%); at h = 2 it meets 0.40 (0.420); at h = 3 it misses 0.25 (0.175). The margin is largest where persistence collapses (0.008 and –0.198 at h = 2 and 3). Peaks, however, are under-predicted at all horizons: the peak bias at the 99th percentile is –0.45, –0.76 and –0.89, worse than persistence at h = 1 (–0.36). The model wins on average error but cuts flood peaks, the most serious weakness for a warning product. Configuration E, the main model from this report on (Section IV-D), keeps h = 1 and raises h = 2 and 3 to 0.499 and 0.236, still short of 0.25 at h = 3.

TABLE II
BASELINES AND LIGHTGBM, TEST PERIOD JULY 2022 – AUGUST 2026. RAW GLOFAS FORECASTS ARE NOT ARCHIVED (III-D); CONFIG. A: REPORT 2 MODEL, L1, SEED 42; CONFIG. E: MAIN MODEL (L1 AT H = 1, L2 AT H = 2–3), MEAN OF THREE SEEDS, RMSE NOT RECORDED (N/R)

Model	NSE h=1	h=2	h=3	RMSE h=1 (m³/s)	h=2	h=3
Persistence	0.542	0.008	–0.198	156.4	230.2	253.1
Climatology	0.097	0.097	0.096	219.6	219.7	219.8
Seasonal naive	–0.434	–0.434	–0.435	276.8	276.8	276.9
ARIMA(2,1,2)	–0.093	–0.093	–0.093	241.6	241.7	241.8
Raw GloFAS forecast	n/a	n/a	n/a	n/a	n/a	n/a
LightGBM, config. A	**0.747**	**0.420**	**0.175**	**116.3**	**176.1**	**210.0**
LightGBM, config. E	**0.747**	**0.499**	**0.236**	**n/r**	**n/r**	**n/r**

B. Risk-level detection

The level-1 classifier detects 20 of the 31 level-1 days one day ahead with 5 false alarms (POD 0.645, FAR 0.200, CSI 0.556; Table III), better than thresholding the regression forecast (CSI 0.394). Skill decays quickly: CSI 0.268 at h = 2

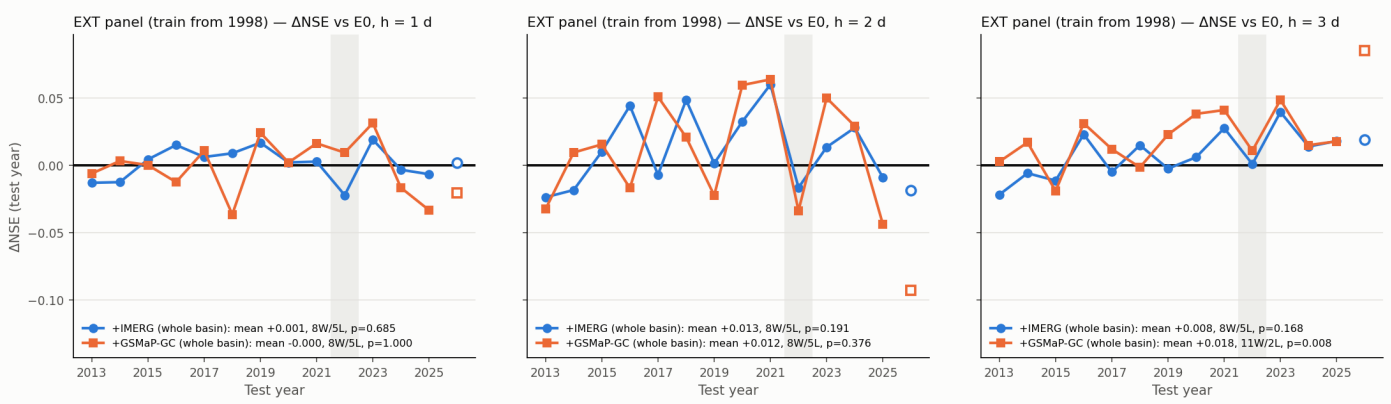


Fig. 2. Satellite rainfall under yearly walk-forward (baseline E0, train from 1998). ΔNSE of each test year for IMERG and gauge-calibrated GSMaP added to E0, ensemble of three seeds; legends give the mean, folds won/lost and Wilcoxon p over 2013–2025. Shaded: GloFAS regime change (07/2022). Hollow markers: January–August 2026, excluded from the summary.

and 0.087 at h = 3. Level 2 has only 12 positive test days; its POD is 0.333 at h = 1 and zero beyond. The acceptance criterion for alert stage II (POD of at least 0.7 one day ahead) can therefore be neither met nor evaluated reliably with this test set.

TABLE III
DEDICATED RISK-LEVEL CLASSIFIERS ON THE TEST PERIOD

Level	h	Positive test days	POD	FAR	CSI	PR-AUC
1 (806 m³/s)	1	31	0.645	0.200	0.556	0.671
	2	31	0.355	0.476	0.268	0.354
	3	31	0.129	0.789	0.087	0.153
2 (1,275 m³/s)	1	12	0.333	0.500	0.250	0.419
	2	12	0.000	1.000	0.000	0.082
	3	12	0.000	1.000	0.000	0.035

C. *Imperfect rainfall (scenario B')*

Scenario B planned to replace observed by archived forecast rainfall, but the archive returned values identical to ERA5 (Report 2). It was replaced by a sensitivity test with multiplicative log-normal noise of growing σ on all rainfall inputs. At σ = 1, NSE falls from 0.742 to 0.683 at h = 1 (0.059 lower), from 0.422 to 0.384 at h = 2, and does not fall at h = 3 (0.167 to 0.172); the noise-free run matches the main model within ±0.01. The model leans on discharge lags, so imperfect rainfall costs little; for the same reason it cannot anticipate rain that has not yet fallen, which caps skill at h = 2–3.

D. *Does more data help? (configurations A–E)*

All five configurations use the per-horizon objective of Section III-C. A longer record (B) is the only change that helps at h = 1 (+0.021; Table IV), and it also helps at h = 2 and 3. Soil moisture (C) and storage proxies (D) give small changes of mixed sign, within about two seed standard deviations: that information is largely carried by discharge lags and the API. Storm, cold-surge and ENSO predictors (E) give the largest gain at h = 2 and 3 (+0.038 and +0.037 over D; at h = 2 from 0.432 for A to 0.499): these drivers announce heavy rain one to three days ahead, which observed rainfall cannot. The seed standard deviation is at most 0.012. No configuration removes the peak bias (–0.42 to –0.89), which points to the target itself (Section IV-I).

TABLE IV
CUMULATIVE ABLATION, SINGLE SPLIT, MEAN OF THREE SEEDS; L1 AT H = 1, L2 AT H = 2–3 FOR EVERY CONFIGURATION

Config.	Added	NSE h=1	h=2	h=3
A	Report 2 panel, train from 2010 (4,550 days)	0.745	0.432	0.166
B	Rainfall from 1997 (9,298 days)	0.766	0.447	0.192
C	+ soil moisture	0.757	0.454	0.208
D	+ reservoir-storage proxies	0.756	0.461	0.199
E	+ storms, cold surges, ENSO	0.747	0.499	0.236

E. *Hourly rainfall (ERA5)*

Hourly ERA5 exists only from 2015, so this test uses its own split (train 2015–2020, validation to June 2022, test from July 2022); its baseline is therefore lower. Nine basin-wide hourly features raise NSE at h = 3 from 0.168 to 0.183, consistently across seeds (+0.016, +0.008, +0.022). At h = 2 the mean gain (0.416 to 0.431) comes almost entirely from one seed (+0.004, +0.041, +0.001). At h = 1 the gain is not intra-day information: daily totals over the DEM sub-basins alone reach 0.726, against 0.723 with hourly features. Hourly features per sub-basin are no better than the basin mean (0.722, 0.429, 0.184): ERA5 is too coarse to separate them.

F. *Satellite rainfall under walk-forward*

On the single split, IMERG appeared to lift E0 at h = 2 (+0.017; +0.007 at h = 3). Across 13 yearly folds the gains are +0.013 and +0.008, with intervals that include zero (Table V, Fig. 2). Gauge-calibrated GSMaP is the only source with a consistent gain on the strong baseline: +0.018 at h = 3, positive in 11 of 13 folds (p = 0.008), and fewer level-1 false alarms at h = 1 (24 to 19 over 105 pooled event days; CSI 0.450 to 0.484). At h = 2 the mean gain (+0.012) is not significant (p = 0.38), and at h = 1 there is none. On the weaker baseline A0 the

same source gains more (+0.032 at h = 2, positive in 13 of 13 folds): satellite rain partly stands in for predictors that E0 already contains.

TABLE V
ΔNSE FROM SATELLITE RAINFALL, 13 YEARLY FOLDS: MEAN, 95% INTERVAL, FOLDS WON/LOST. BASELINE NSE: E0 0.761/0.437/0.206, A0 0.744/0.336/0.121

Config. (vs. own baseline)	h = 1	h = 2	h = 3
E0 + GSMaP-GC	–0.001 (8/5)	+0.012 [–0.008; +0.031] 8/5, p 0.38	+0.018 [+0.008; +0.028] 11/2, p 0.008
E0 + IMERG	+0.001 (8/5)	+0.013 [–0.001; +0.027] 8/5, p 0.19	+0.008 [–0.001; +0.017] 8/5, p 0.17
A0 + GSMaP-GC	+0.006 (9/4)	+0.032 [+0.023; +0.041] 13/0	+0.025 [+0.013; +0.038] 12/1
E0 + GSMaP-GC, same day (latency run)	+0.004 (10/3)	+0.018 (9/4)	+0.026 [+0.015; +0.036] 11/2, p 0.002
E0 + GSMaP-GC, lagged one day	+0.001 (8/5)	+0.004 (7/6)	–0.001 [–0.007; +0.005] 8/5, p 0.839

Latency. On 20 October 2026 the newest GSMaP image on Earth Engine was about 27 h old (IMERG about 38 h). A forecast issued near 05:00 local time therefore has satellite rain only up to day t – 1. Re-running the walk-forward with GSMaP-GC features lagged by one day removes the gain: –0.001 [–0.007; +0.005] at h = 3 (8/5 folds, p = 0.839), against +0.026 (p = 0.002) with same-day features in the same run. The benefit comes entirely from rain on the issue day itself. *The satellite gain is a reanalysis result; it is not yet an operational one.*

G. Low-latency satellite rainfall (GSMaP_NRT)

JAXA distributes a gauge-adjusted near-real-time GSMaP product published about 4.7 h after observation and reprocessed back to 1998. It is a different product from the standard GSMaP on Earth Engine (hourly correlation 0.6–0.8), so it has to be tested by retraining on its own history rather than by substitution. With the forecast job moved after 07:00 local time, this product would supply day-t rain operationally. Table VI gives the walk-forward result.

TABLE VI
ΔNSE FROM GSMAP_NRT AND STANDARD GSMAP-GC, YEARLY WALK-FORWARD

Config. (vs. own baseline)	h = 1	h = 2	h = 3
A0+GSMaP-GC (EE)	+0.006 [+0.000; +0.011] 9/4, p 0.068	+0.032 [+0.023; +0.041] 13/0, p 0.000	+0.025 [+0.013; +0.038] 12/1, p 0.005
A0+GSMaP_NRT	+0.009 [+0.003; +0.014] 10/3, p 0.008	+0.019 [+0.008; +0.031] 11/2, p 0.006	+0.012 [+0.005; +0.019] 12/1, p 0.005
E0+GSMaP-GC (EE)	–0.000 [–0.011; +0.010] 8/5, p 1.000	+0.012 [–0.008; +0.031] 8/5, p 0.376	+0.018 [+0.008; +0.028] 11/2, p 0.008
E0+GSMaP_NRT	–0.002 [–0.007; +0.003] 4/9, p 0.273	+0.003 [–0.009; +0.017] 6/7, p 0.735	–0.001 [–0.009; +0.008] 5/8, p 0.542

[Interpretation pending team review] *Numbers above are computed from* `reports/wf_sat_nrt.csv` *at build time; the conclusion sentence must be written after the team has checked them.*

H. Deep learning versus LightGBM

On the single split some of the 8 DL configurations beat LightGBM at h = 2 and 3 (best: a peak-weighted MLP, 0.490 against 0.420 at h = 2), but the LSTM configuration selected on validation lost at all three horizons (0.709, 0.404, 0.171 against 0.747, 0.420, 0.175); the better configurations were identified only after seeing the test set. Under the yearly walk-forward, with every model on the Report 2 panel, no DL model beats LightGBM consistently (Table VII, Fig. 3). The LSTM with MSE loss is level at h = 1 (+0.021, interval includes zero) and slightly worse at h = 2–3; the peak-weighted LSTM loses clearly at h = 3 (–0.229, p = 0.033). The single-split advantage at h = 2–3 appears only in 2022–2025, the years that split happened to test; in 2013–2021 every DL model loses on average at h = 2 and 3.

TABLE VII
DEEP LEARNING VERSUS LIGHTGBM, 13 YEARLY FOLDS: ΔNSE MEAN, 95% INTERVAL, FOLDS WON/LOST; POOLED CSI

Model	h = 1	h = 2	h = 3	CSI level 1, h = 1
LightGBM (NSE)	0.743	0.334	0.118	0.44
LSTM L=30, MSE	+0.021 [–0.014; +0.058] 8/5	–0.025 [–0.117; +0.032] 7/6	–0.026 [–0.101; +0.021] 7/6	0.49
LSTM L=30, peak-weighted	–0.007 [–0.065; +0.037] 8/5	–0.107 [–0.239; –0.008] 5/8	–0.229 [–0.411; –0.073] 5/8	0.49
MLP on lags, peak-weighted	+0.000 [–0.031; +0.037] 5/8	–0.065 [–0.142; +0.002] 5/8	–0.118 [–0.255; –0.015] 6/7	0.49

DL has two genuine strengths at h = 1: it detects more level-1 days (POD 0.59–0.63 against 0.51; pooled CSI 0.49

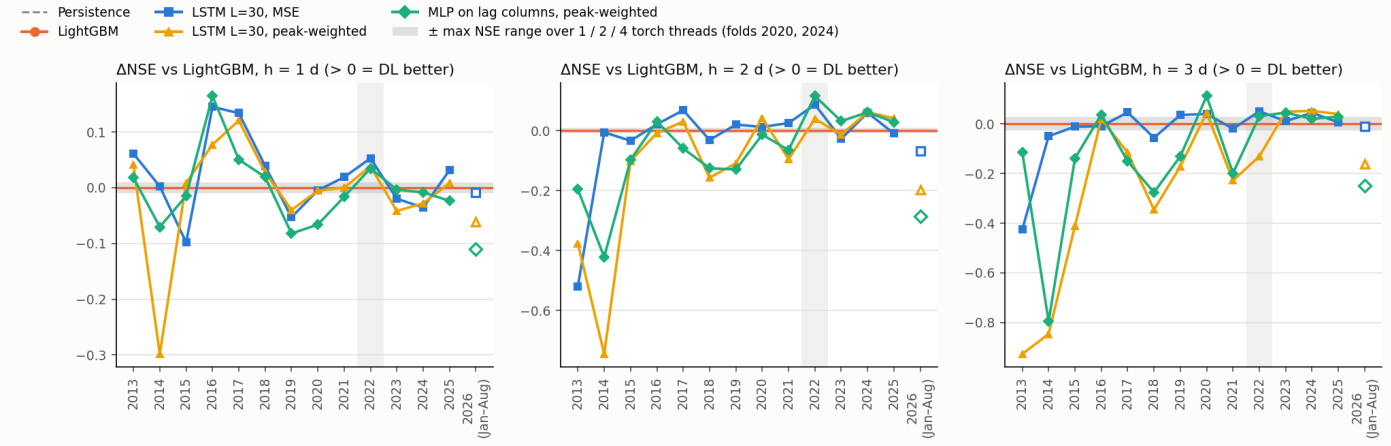


Fig. 3. Deep learning against LightGBM under the same walk-forward (panel from 2010): ΔNSE per test year, positive when DL is better. The grey band is the largest change caused by the torch thread count (folds 2020 and 2024).

against 0.44) and cuts peaks less (peak bias –0.21 to –0.26 against –0.38). It is also far less stable: the seed-to-seed spread is 0.03–0.20 NSE against at most 0.01 for LightGBM, and the torch thread count moves the MLP by up to 0.028 while leaving the LSTM unchanged. LightGBM remains the main model.

I. Reservoirs: is the dam signal in GloFAS?

If GloFAS v4 represented the Ta Trach (646 million m³) and Binh Dien (424 million m³) reservoirs, the simulated series after both dams were operating (2014–2026) should show higher dry-season flow, lower flashiness and a weaker storm response than before them (1998–2008). None of the four indices changes significantly (Table VIII), and the three dam-sensitive ones move the wrong way or not at all. The target behaves as an unregulated river, so reservoir effects cannot be learned from it (R1).

TABLE VIII
GLOFAS BEFORE (1998–2008, 11 YEARS) AND AFTER (2014–2026, 13 YEARS) THE DAMS: MEDIANS, MANN–WHITNEY p (STORM RESPONSE: 25 AND 37 EVENTS)

Index	Before	After	Change	p
30-day low flow (m³/s)	29.7	26.5	–10.7%	0.42
Runoff ratio	0.857	0.875	+2.2%	0.91
Flashiness (Richards–Baker)	0.435	0.441	+1.3%	0.64
Storm response	5.80	5.98	+3.2%	0.64

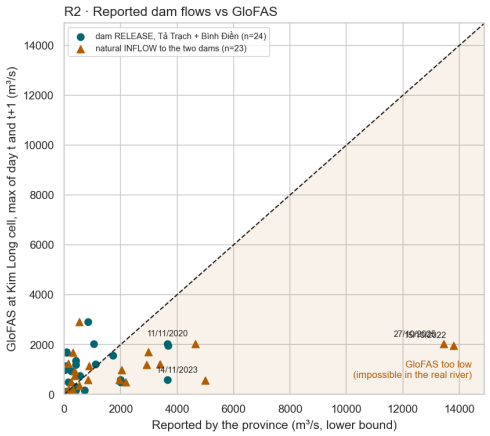


Fig. 4. Reported dam releases and natural inflows (lower bounds) against GloFAS at the Kim Long cell. Points below the diagonal are physically impossible for the real river.

Bulletins give two conservative lower bounds for the flow reaching Kim Long: the smallest release reported for each reservoir on a day, and the reported natural inflow to the two reservoirs, whose catchments (1,232 km²) make up most of the basin. GloFAS (the larger of days t and t + 1) is below the release bound on 11 of 24 days and below the inflow bound on 14 of 23 (Fig. 4, R2). On 15 October 2022 the dams released at least 3,672 m³/s with an inflow of 13,787 m³/s, while GloFAS gave 1,967 m³/s; on 27 October 2025 the figures were 3,669, 13,433 and 2,022 m³/s. Bulletins report instantaneous values and GloFAS daily means, which explains part of the gap but not several-fold differences. The dominant problem is not reservoir operation but that GloFAS under-simulates the largest floods, consistent with ERA5 under-estimating extreme orographic rain. This supports relative, percentile-based risk levels and motivates the floor rule of Section III-H.

V. EXPLAINABILITY

Split-gain importance is a poor guide for this model: at h = 2 the two seasonal encodings rank first and second by gain, yet removing them in an earlier ablation did not lower NSE at any horizon, because the same information is carried by discharge and rainfall lags. SHAP values attribute each individual forecast to its inputs and are a standard explanation tool for tree ensembles in streamflow forecasting [9]; they are used here to check whether the main model relies on hydrologically plausible signals.

A. Set-up

Tree SHAP values were computed for configuration E (88 features), trained on train plus validation with seed 42 and explained on the test period, in units of log(1 + Q) so that the base value plus the contributions equals the forecast. The run reproduces the ablation exactly (NSE 0.748, 0.503, 0.231). Correlated inputs (discharge and its lags; the three rainfall bands) share credit arbitrarily, so results are read by mechanism group (Table IX); recomputing the shares with seeds 7 and 2026 changes them by at most 3.6 percentage points.

TABLE IX
SHARE OF GROUPED MEAN |SHAP| (%), CONFIGURATION E, SEED 42

Group	All test days h=1	h=2	h=3	Level-1 days h=1	h=2	h=3
Lagged discharge	58.4	26.3	22.8	57.4	20.7	21.7
Rain, up to 7 days	33.3	36.5	22.5	35.0	44.5	23.3
Long rain / reservoir proxy	2.2	5.1	7.8	1.5	3.5	6.2
Soil moisture and API	1.8	5.6	4.7	2.6	5.9	3.8
Tropical cyclone	0.9	3.5	4.4	0.7	3.5	6.0
Pressure, wind, humidity	2.3	12.7	16.3	1.8	14.1	16.8
ENSO	0.5	3.0	5.9	0.3	1.9	5.2
Calendar	0.7	7.2	15.6	0.7	6.0	16.9

B. What the model relies on

Today's discharge dominates at h = 1 (37.4% of mean |SHAP|), followed by upper-basin (11.2%) and basin-mean rainfall (9.5%). With lead time the weight moves from lagged discharge (58.4% to 22.8% by group) to rainfall, which peaks at h = 2 (36.5%; basin rainfall is the first feature), and then to large-scale drivers: the pressure, wind and humidity group grows from 2.3% to 16.3%, and the north–south pressure gradient that marks cold surges becomes the third feature at h = 3 (5.2%), behind discharge (6.5%) and basin rainfall (6.1%). This is the mechanism behind the +0.038 gain of configuration E at h = 2. The direction is physically right for the core inputs: the Spearman correlation between a feature and its own SHAP value is +1.00 for discharge at h = 1 and +0.89 to +0.95 for the pressure gradient.

C. Where the model is not plausible

- **Soil moisture acts in reverse.** For the 7–28 cm ERA5-Land layer the correlation is –0.12, –0.25 and –0.27: wetter soil lowers the forecast, and the soil and API group is negative on level-1 days at every horizon (–0.06, –0.13, –0.04 in log units). Soil moisture largely duplicates the wetness already contained in today's discharge, so the model uses it as a correction term. It must not be cited as a flood mechanism.
- **The full-reservoir proxy is not supported.** Rain accumulated since 1 September has correlations of –0.16, +0.02 and +0.02, consistent with R1: the target contains no reservoir.
- **Climatology at h = 3.** The calendar group carries 15.6% at h = 3, almost as much as the pressure and wind group (16.3%): when the dynamical signal is weak the model partly falls back on the season.

D. Why peaks are cut

For the forecast issued on 15 October 2022 (Fig. 5), discharge and rainfall push the forecast strongly upward, yet it reaches only 862 m³/s against a GloFAS peak of 1,967 m³/s. The limit is structural rather than a missing feature: only 12 of the 9,283 training days exceed that peak, each leaf must hold at least 30 training days, and the L1 loss used at h = 1 targets the conditional median of log discharge. Moreover, every training day with discharge above 2,000 m³/s receded the next day, so the model predicts recession even while the river is still rising: on 30 October 2025 it gave 1,727 m³/s against 2,903 m³/s, the largest flow of the test period. More predictors will not fix this; in the yearly walk-forward a smaller minimum leaf size did not help and an upper-quantile objective traded NSE for detection.

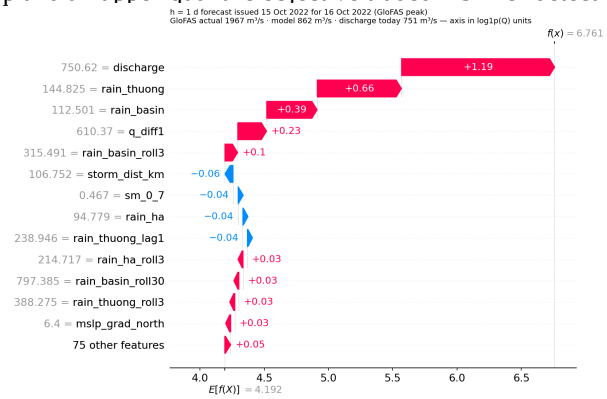


Fig. 5. SHAP waterfall (log(1 + Q) units) for the forecast issued on 15 October 2022 for the GloFAS peak of the next day: discharge and rainfall push upward, but the forecast stays far below the peak.

VI. DISCUSSION AND LIMITATIONS

- **The target is simulated.** Every score measures agreement with GloFAS, not with the river. LightGBM cuts peaks relative to GloFAS, and GloFAS itself under-simulates the largest floods (Section IV-I), so true peaks are under-estimated further. An upper-quantile objective tested in the yearly walk-forward detected more level-1 days but

lowered NSE and still fell below most peaks at h = 2–3, so it was not adopted.

- **Regime change in July 2022.** The test period consists of archived forecasts. Report 2 found no systematic shift in the rainfall response (Mann–Whitney p = 0.581 in the heavy-rain tail), but that test assumes ERA5 is right, and the archived regime has too few folds to be tested alone. GloFAS v5 adds large reservoirs; Ta Trach and Binh Dien meet its storage and area criteria, so the target may change when Open-Meteo moves to v5, and R1 must then be repeated.
- **Satellite latency.** The h = 3 gain from GSMaP-GC exists only when same-day rain is available. Through Earth Engine it is not; the operational value depends on the GSMaP_NRT test (Section IV-G).
- **No hourly target.** Finer and hourly inputs were tested, but the target is still daily GloFAS discharge on a ~5 km grid. An hourly target would need observed hourly water levels at Kim Long from the national hydro-meteorological service; the team decided not to request these data, so hourly forecasting of the river itself is out of scope.
- **Rare events and labels.** Only level 1 has enough test events (31 days). Risk levels are relative to GloFAS and are not official alert stages, because discharge explains little of the water level ($R^2 = 0.18$).
- **Resolution of comparisons.** NSE differences below about 0.01 are numerical noise; differences between configurations are reported with fold-level intervals, and statements rest on those intervals, not on means.

VII. CONCLUSION AND PLAN FOR REPORT 4

LightGBM on open data forecasts GloFAS discharge at Kim Long with clear skill over all baselines (NSE 0.747, 0.420, 0.175) and detects level-1 risk one day ahead with CSI 0.556, but cuts peaks and loses most skill beyond one day. Of the factors raised in the review, reservoir releases cannot be learned from a target that does not contain them and are handled by a floor rule; large-scale weather predictors help at h = 2; satellite and hourly rainfall help at h = 3 only in hindsight; and deep learning does not beat LightGBM once evaluated across years. Report 4 will:

- run the GSMaP_NRT walk-forward and, if the gain holds, move the daily job after 07:00 so that day-t satellite rain is available;
- compare the model with the raw GloFAS forecasts logged since 15 September 2026, and report the event-based scores of the 2026 flood season;
- test, on validation folds only, the removal of soil moisture and of the reservoir proxy that SHAP flags as implausible, and address peak cutting with quantile regression or a smaller minimum leaf size, and with a cost-based warning threshold;
- present the Spark pipeline honestly: on a two-worker cluster it reproduces the Polars ETL exactly (6,544,896 hourly rows to 272,704 cell-days) but takes 40.5–41.2 s against 3.4–3.7 s for Polars on one machine; MLlib models (GBT 0.699, random forest 0.687 at h = 1) do not beat LightGBM; and
- report the walk-forward as the main comparison for every model, with the single split as one case among thirteen.

REFERENCES

[1] Open-Meteo, “Flood API and Historical Weather API,” 2026. [Online]. Available: https://open-meteo.com/en/docs

[2] S. Harrigan *et al.*, “Daily ensemble river discharge reforecasts and real-time forecasts from the operational Global Flood Awareness System,” *Hydrol. Earth Syst. Sci.*, vol. 27, no. 1, pp. 1–19, 2023.

[3] H. Hersbach *et al.*, “The ERA5 global reanalysis,” *Q. J. R. Meteorol. Soc.*, vol. 146, no. 730, pp. 1999–2049, 2020.

[4] Prime Minister of Vietnam, “Decision 05/2020/QD-TTg on water levels corresponding to flood alert levels,” *Official Gazette*, no. 179+180, Feb. 2020, Annex I, row 135. [in Vietnamese]

[5] H. S. Nguyen *et al.*, “Synoptic patterns causing heavy rain and floods in Thua Thien Hue province in 2020,” *Hue Univ. J. Sci.: Earth Sci. Environ.*, vol. 131, no. 4A, pp. 149–162, 2022. [in Vietnamese]

[6] G. Nearing *et al.*, “Global prediction of extreme floods in ungauged watersheds,” *Nature*, vol. 627, pp. 559–563, 2024.

[7] V. Nourani, S. Kheirieh, S. A. Kantoush and J. J. Huang, “Bias correction of Global Flood Awareness System (GloFAS) data for multi-station river flow prediction by ensemble modelling,” *Eng. Appl. Comput. Fluid Mech.*, vol. 20, no. 1, 2665857, 2026.

[8] T. Honcharenko *et al.*, “Explainable deep ensemble bias correction of GloFAS-ERA5 streamflow across snow-influenced transboundary basins of Central Asia,” *Water*, vol. 18, no. 16, 2055, 2026.

[9] S. Gnjato, I. Lescesen, X. Zhou and D. Dukanovic, “Explainable machine learning for streamflow forecasting: application to the Bosna River basin,” *Water*, vol. 18, no. 10, 1226, 2026.

[10] J. H. Ougahi and J. S. Rowan, “Investigating deep learning knowledge transfer in streamflow prediction from global to local catchment,” *Water Resour. Res.*, 2026, doi: 10.1029/2025WR041194.

[11] Prime Minister of Vietnam, “Decision 1606/QD-TTg on the inter-reservoir operating procedure for the Huong River basin,” Nov. 2019. [Online]. Available: https://datafiles.chinhphu.vn/cpp/files/vbpq/2019/11/1606.signed.pdf [in Vietnamese]

[12] UNESCO-IHP, *Catalogue of Rivers for Southeast Asia and the Pacific*, vol. V, entry Viet Nam-8 (Huong River).

[13] NASA, “GPM IMERG V07, Earth Engine collection NASA/GPM_L3/IMERG_V07.”

[14] JAXA, “GSMaP v8 operational, Earth Engine collection JAXA/GPM_L3/GSMaP/v8/operational.”

[15] NOAA NCEI, “International Best Track Archive for Climate Stewardship (IBTrACS) v04r01.” [Online]. Available: https://www.ncei.noaa.gov/data/international-best-track-archive-for-climate-stewardship-ibtracs/

[16] NOAA CPC, “Oceanic Niño Index.” [Online]. Available: https://www.cpc.ncep.noaa.gov/data/indices/oni.ascii.txt

[17] ESA/Copernicus, “Copernicus DEM GLO-30.” [Online]. Available: https://copernicus-dem-30m.s3.amazonaws.com/